

**JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
SECTOR-128, NOIDA.**

DBMS(24B15CS213)

B.TECH. (III SEMESTER)

PROJECT BASED LEARNING

Movie Production & Film Industry Database Management System



GROUP DETAILS

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TITLE OF THE PROJECT

Movie Production & Film Industry Database Management System

INTRODUCTION

The film industry is a complex ecosystem that involves multiple stakeholders such as **actors, producers, directors, technicians, managers, studios, OTT platforms, and award organizations**. Managing this data efficiently requires a well-structured **Database Management System (DBMS)**.

This project aims to design and implement a **Movie Production & Film Industry DBMS** that supports:

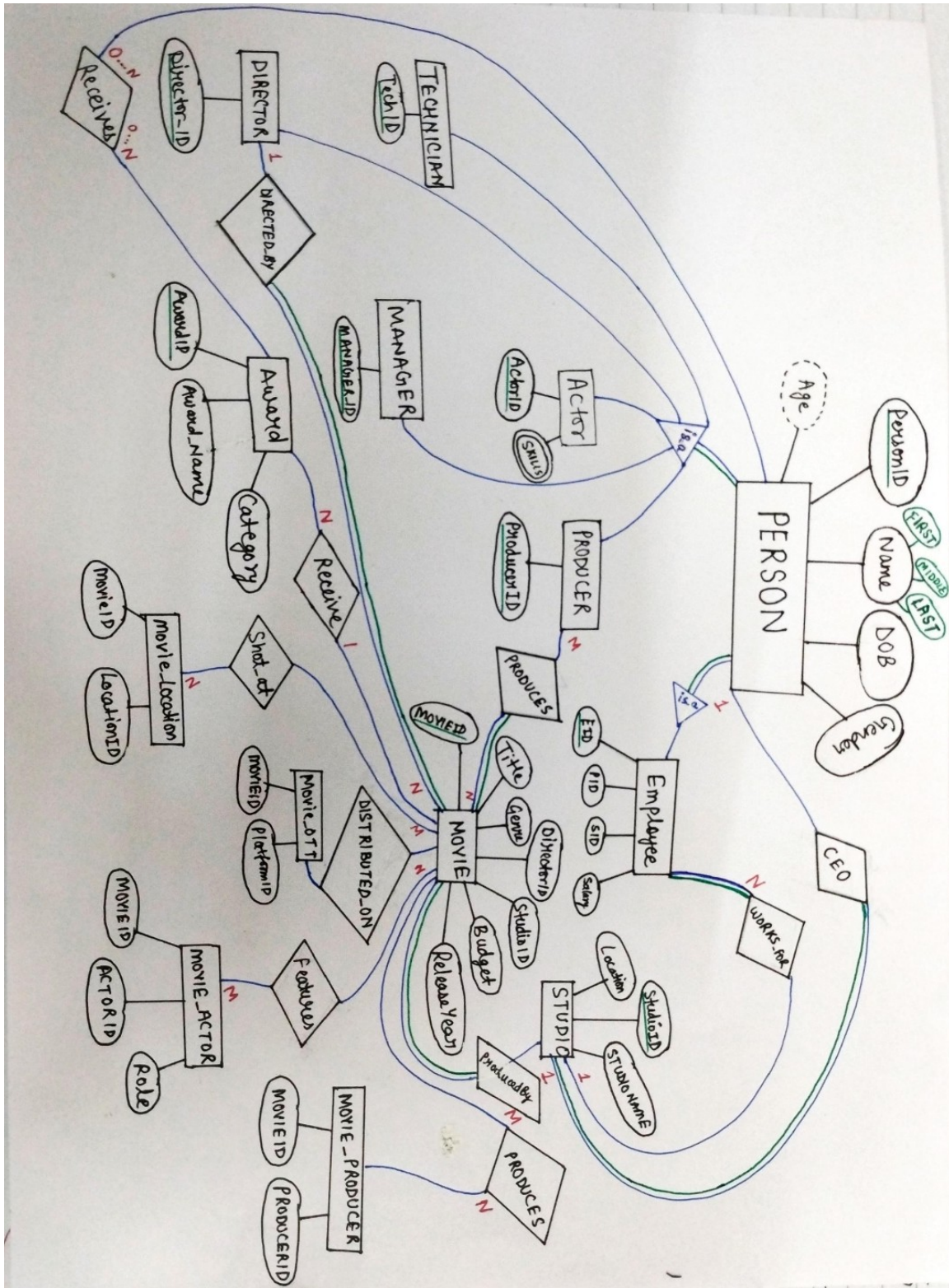
- **Movie lifecycle management** – from production (studio, budget, shooting location) to distribution (OTT rights, box office).
- **Cast and crew handling** – managing actors, directors, producers, employees, and their roles in different movies.
- **Award management** – tracking awards received by movies or people across different years.
- **Queries & Analytics** – retrieving information such as top actors by earnings, highest-grossing directors, or movies distributed on OTT platforms.
- **PL/SQL Features** – stored procedures for payroll, triggers for automatic budget validation, cursors for award listing, and transaction management for movie insertion.
- **Database Security** – enforcing role-based access (e.g., only producers update budgets, only admin insert awards).

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This system provides a **real-world applicable model** and also covers all DBMS syllabus requirements including **ER design, normalization, SQL queries, relational model, PL/SQL, transactions, and security**.

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ER DIAGRAM



Schema with Constraints

```
Person(
    PersonID INT PRIMARY KEY,
    Name VARCHAR(150),
    FirstName VARCHAR(50),
    MiddleName VARCHAR(50),
    LastName VARCHAR(50),
    DOB DATE,
    Gender CHAR(1)
    -- Age is derived from DOB → not stored in schema
);

Actor(
    ActorID INT PRIMARY KEY,
    PersonID INT UNIQUE,
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)
);

Actor_Skills(
    ActorID INT,
    Skill VARCHAR(100),
    PRIMARY KEY(ActorID, Skill),
    FOREIGN KEY (ActorID) REFERENCES Actor(ActorID)
);

Producer(
    ProducerID INT PRIMARY KEY,
    PersonID INT UNIQUE,
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)
);

Director(
    DirectorID INT PRIMARY KEY,
    PersonID INT UNIQUE,
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)
);

Technician(
    TechID INT PRIMARY KEY,
    PersonID INT UNIQUE,
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)
);

Manager(
    ManagerID INT PRIMARY KEY,
    PersonID INT UNIQUE,
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)
);

Employee(
    EmployeeID INT PRIMARY KEY,
    PersonID INT,
    StudioID INT,
    Salary DECIMAL(12,2),
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID),
    FOREIGN KEY (StudioID) REFERENCES Studio(StudioID)
);

Studio(
    StudioID INT PRIMARY KEY,
    StudioName VARCHAR(200),
    Location VARCHAR(100),
    CEO INT UNIQUE, -- reference to Employee
```

```

    FOREIGN KEY (CEO) REFERENCES Employee(EmployeeID)
);
Movie(
    MovieID INT PRIMARY KEY,
    Title VARCHAR(200),
    ReleaseYear INT,
    Genre VARCHAR(100),
    StudioID INT,
    DirectorID INT,
    Budget DECIMAL(12,2),
    FOREIGN KEY (StudioID) REFERENCES Studio(StudioID),
    FOREIGN KEY (DirectorID) REFERENCES Director(DirectorID)
);
Movie_Actor(
    MovieID INT,
    ActorID INT,
    Role VARCHAR(100),
    Payment DECIMAL(12,2),
    PRIMARY KEY(MovieID, ActorID),
    FOREIGN KEY (MovieID) REFERENCES Movie(MovieID),
    FOREIGN KEY (ActorID) REFERENCES Actor(ActorID)
);
Movie_Producer(
    MovieID INT,
    ProducerID INT,
    PRIMARY KEY(MovieID, ProducerID),
    FOREIGN KEY (MovieID) REFERENCES Movie(MovieID),
    FOREIGN KEY (ProducerID) REFERENCES Producer(ProducerID)
);
Movie_Location(
    MovieID INT,
    LocationID INT,
    PRIMARY KEY(MovieID, LocationID),
    FOREIGN KEY (MovieID) REFERENCES Movie(MovieID),
);
Movie_OTT(
    MovieID INT,
    PlatformID INT,
    StartDate DATE,
    EndDate DATE,
    RightsCost DECIMAL(12,2),
    PRIMARY KEY(MovieID, PlatformID),
    FOREIGN KEY (MovieID) REFERENCES Movie(MovieID),
);
Award(
    AwardID INT PRIMARY KEY,
    AwardName VARCHAR(100),
    Category VARCHAR(100)
);

```